



## ARTIFICIAL INTELLIGENCE IN ASSISTANCE FOR INDIVIDUALS WITH HEARING, VISUAL, AND MOTOR DISABILITIES

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**Summary** - This article reviews the use of AI in assistive technologies for hearing, visual, and motor disabilities, focusing on computer vision, neural networks, and NLP. It highlights advances in accessibility, autonomy, and challenges such as privacy concerns.

**Keywords:** Artificial Intelligence; Assistive Technology; Disabilities; Healthcare Technology; Accessibility.

### INTRODUCTION

Artificial Intelligence (AI) has played an important role in the development of assistive technologies aimed at individuals with hearing, visual, and motor disabilities, with the goal of improving their quality of life by promoting greater autonomy and social inclusion. It is estimated that by 2050, approximately 2.5 billion people will have some degree of hearing loss, of which 700 million will require hearing rehabilitation [1]. Similarly, visual impairment, whether congenital or acquired, significantly affects learning, work, and social interaction, thereby reducing quality of life [2]. AI-based assistive technologies have provided innovative solutions to improve mobility and interaction with the environment.

In the case of motor disabilities, a report by the World Health Organization (WHO) and the United Nations International Children's Emergency Fund (UNICEF) states that over 2.5 billion people currently require assistive technology products, and this number is expected to reach 3.5 billion by 2050, driven by global population aging and the rise of chronic diseases [3]. In Brazil, approximately 3.4% of the population experiences mobility difficulties, and 1.4% faces challenges in tasks requiring manual dexterity, such as grasping small objects [4].

The objective of this work is to review the main AI techniques applied in the development of assistive technologies for individuals with hearing, visual, and motor disabilities. The literature review aims to identify, select, and synthesize the available scientific evidence, following a structured approach that ensures the reproducibility of the investigation [5]. The document is organized as follows: the next section addresses the methodology used in the literature review, followed by the presentation and discussion of the results.

### DEVELOPMENT

To conduct this literature review, a research approach was developed that followed specific guidelines to ensure the organization and reproducibility of the results obtained from scientific databases, as recommended by Kitchenham and Charters [5]. This approach included the research objective, the formulation of a central question, the definition of search strings, the selection of research databases to be used, and the criteria for inclusion and exclusion of studies.

The research question was formulated with the objective of identifying which assistive technologies

based on Artificial Intelligence (AI) have been used to improve the quality of life and autonomy of people with hearing, visual, and motor disabilities. The main question that guided the investigation was: "Which AI techniques applied in assistive technologies contribute to improving the quality of life and autonomy of individuals with hearing, visual, and motor disabilities?"

Based on this question, a specific search string was created that combined terms related to AI, assistive technologies, and hearing, visual, and motor disabilities. The search string used was: "Artificial Intelligence" AND ("Assistive Technology" OR "Assistive Devices") AND ("Hearing Impairment" OR "Visual Impairment" OR "Motor Impairment") AND Accessibility. This search was conducted in renowned academic databases, including ScienceDirect, Scopus, and IEEE Xplore, focusing exclusively on research articles.

The article selection process was conducted according to inclusion and exclusion criteria. Studies that explicitly addressed the application of AI in assistive technologies aimed at improving the independence and social inclusion of people with disabilities were included. Only publications from 2019 to 2024 were considered, with the aim of capturing the most recent innovations. Studies that did not mention AI or focused on areas unrelated to hearing, visual, or motor disabilities were excluded, as well as purely theoretical articles without practical applicability. The selected studies were categorized by the type of disability they addressed and the specific AI techniques applied, such as computer vision, neural networks, and natural language processing.

To assist in organizing and conducting the review, the Rayyan (<https://rayyan.ai/>) tool was used, which facilitated the process of screening and analyzing the articles.

### RESULTS AND DISCUSSIONS

After applying the search string in the academic databases ScienceDirect, Scopus, and IEEE Xplore, 70 articles were initially identified. Of these, 35 articles were selected for detailed analysis, and two duplicate articles were removed, resulting in 33 final articles considered for the review.

Regarding the distribution of articles by database, it was noted that the majority of articles were retrieved from the ScienceDirect database (39 articles), followed by IEEE Xplore (21 articles) and Scopus (10 articles).

As for the distribution of published articles by year, there has been a significant increase in the



number of publications in 2023 and 2024, as illustrated in Figure 1.

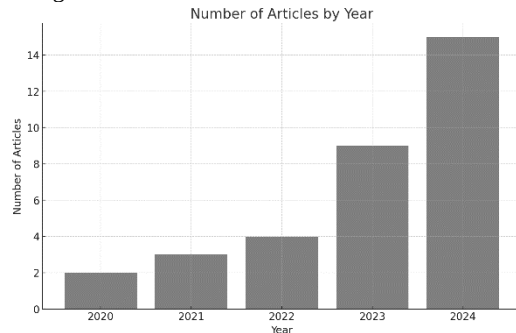


Figure 1 - Number of Articles Published by Year

Moreover, 73% of the articles focus on visual disabilities, 18% address hearing disabilities, and 9% focus on motor disabilities, as shown in Figure 2.

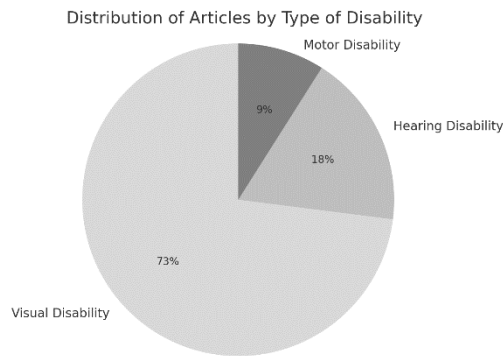


Figure 2 - Distribution of Articles by Type of Disability

During the selection phase, inclusion and exclusion criteria were applied, resulting in the analysis of 33 articles. As summarized in Table 1, these articles cover a variety of techniques related to the use of AI to improve accessibility and independence for individuals with visual, hearing, and motor disabilities. It is noteworthy that some articles fall into more than one technology category. For example, articles [28], [29], and [32] are related to the development of object detection systems, facial recognition, and assisted mobility devices, demonstrating the overlap of AI techniques in addressing different challenges faced by individuals with disabilities.

TABLE 1 - AI Techniques Applied in Assistive Technologies

| Type of Disability | Assistive Technology                                     | AI Techniques                   | References                                    |
|--------------------|----------------------------------------------------------|---------------------------------|-----------------------------------------------|
| Visual             | Systems for image and graphics description and reading   | Computer Vision, Deep Learning  | [6], [18], [23], [26], [31], [33], [37]       |
|                    | Applications for object detection and facial recognition | Computer Vision, Deep Learning  | [8], [19], [24], [27], [28], [29], [32], [36] |
|                    | Navigation and orientation systems                       | AI, Natural Language Processing | [10], [11], [16], [25], [34]                  |

|         |                                                       |                                              |                                    |
|---------|-------------------------------------------------------|----------------------------------------------|------------------------------------|
| Hearing | Smart canes and assisted mobility devices             | Sensors, Computer Vision                     | [11], [19], [28], [29], [32], [36] |
|         | Applications for text and currency recognition        | Computer Vision, Deep Learning               | [24], [29], [30], [35]             |
|         | Braille translators and Braille learning              | AI, Deep Learning                            | [30]                               |
|         | Body tracking systems for mobility assessment         | Computer Vision                              | [17]                               |
|         | Virtual assistants and voice interfaces               | AI, Natural Language Processing              | [20], [23], [25], [26], [29]       |
|         | Text-to-sign language translation systems             | Deep Learning, Neural Networks               | [9], [13], [14], [22], [38]        |
|         | Interaction with intelligent virtual assistants       | AI, Natural Language Processing              | [21]                               |
|         | EEG-based brain-computer interfaces                   | Machine Learning, Fractal Analysis           | [12]                               |
|         | Cursor control using facial recognition               | Convolutional Neural Networks, Deep Learning | [7]                                |
|         | Communication for total paralysis using eye movements | Convolutional Neural Networks, Deep Learning | [15]                               |

The reviewed articles addressing visual impairments primarily apply Computer Vision and Deep Learning techniques. The study [6] uses Computer Vision to interpret charts, generating textual summaries that are understandable for visually impaired individuals. Other works, such as [8], explore real-time facial recognition, emphasizing improved mobility and user safety through accurate detection of objects and faces. Article [24] focuses on currency and text recognition to promote financial autonomy.

Additionally, some approaches focus more on autonomous navigation and orientation, like the system based on a Conversational Interface described in [10], which provides safe support for urban mobility. The integration of navigation systems with natural language processing was also identified, such as in the study [25], which aims to enhance the interaction between the user and assistive systems for the visually impaired.

Studies related to hearing impairments predominantly apply deep neural networks and machine learning techniques. Article [9] presents a system for translating Arabic sign language to text using Convolutional Neural Networks (CNN), showing improvements over previous rule-based systems. Similarly, [22] proposes a real-time sign language recognition approach incorporating recurrent networks to enhance detection accuracy. The articles also explore the interaction between virtual assistants and hearing-impaired individuals. In the study [21], the use of intelligent personal assistants like Alexa is



discussed, which incorporates accessibility options such as sign language translation, demonstrating advancements in digital inclusion for this audience.

For motor impairments, AI systems focus primarily on brain-computer interfaces (BCIs) and facial recognition control. Article [12] explores the use of Fractal Analysis and Machine Learning to improve the accuracy of motor task classification in BCIs, demonstrating the effectiveness of new complexity metrics to enhance the control of assistive devices. Another study [7] discusses the use of convolutional neural networks for cursor control through facial recognition, increasing the independence of users with severe paralysis.

The results of the reviewed studies indicate that AI techniques, especially Computer Vision, Deep Learning, and Natural Language Processing, have been frequently applied in assistive technologies. However, the reported challenges include the need for greater accessibility to devices and the difficulty of integrating these technologies into users' daily lives. Moreover, some studies highlight concerns related to privacy, such as in the case of camera-based systems for visually impaired individuals [20], emphasizing the need for solutions that balance usability and security.

For future work, the review is expected to be refined by including new criteria and expanding the scope to cover studies that utilize robotics and generative AI techniques, aiming to capture a broader range of solutions and technological perspectives in the field of assistive technologies.

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